India–Japan Trade and Technology Partnerships: Building Sustainable Startup Ecosystems for Viksit Bharat @ 2047

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Abstract:

India's Viksit Bharat @ 2047 vision positions innovation, trade integration, and technology-driven entrepreneurship as central to its development pathway. Japan, as a longstanding economic and technological partner, contributes through bilateral trade, FDI, ODA, and collaborative technology programs. Yet, limited empirical evidence explains how these macro-level linkages translate into micro-level outcomes for India's startup ecosystem. This study addresses that gap using an Augmented Gravity Model (2010–2023) and a Difference-in-Differences design. The gravity estimations show strong explanatory power, with India's GDP expansion and technology spillovers significantly enhancing bilateral trade, while Japanese FDI indicates a substitution effect through local production. The DiD analysis reveals substantial policy impacts: semiconductor startups increased by about 189 units ($p = 0.003$), and green hydrogen startups grew by nearly 42 units ($p = 0.003$) relative to their controls. Overall, the findings demonstrate that India–Japan technology partnerships and targeted industrial missions meaningfully accelerate innovation and support sustainable startup growth.

Introduction

India's ambition to achieve developed-nation status under the *Viksit Bharat @ 2047* vision has placed innovation, technology, and entrepreneurship at the centre of its economic strategy. The transformation of India's start-up landscape reflects this shift: what was once a small metropolitan cluster has expanded into a nationwide ecosystem comprising **159,157 DPIIT-recognised start-ups by early 2025**, up from fewer than 1,000 in 2016, generating **over 1.66 million direct jobs**. India now stands as the **world's third-largest startup ecosystem** (Ministry of Commerce & Industry).
Parallel to this domestic expansion, India's economic engagement with Japan has strengthened considerably. In **2023–24**, bilateral trade reached **US$ 22.85 billion**, with exports at **US$ 5.15 billion** and imports at **US$ 17.69 billion** (JBIC Survey). Japan is also one of India's most influential investment partners, contributing **US$ 43.11 billion**

Review of Literature

Research consistently shows that international trade and technology flows strongly influence innovation

performance. Melitz and Redding (2021) argue that trade expands markets, intensifies competition, and enables global knowledge spillovers, while Akçigit and Melitz (2022) highlight supply-chain linkages as catalysts for firm-level innovation. FDI studies similarly identify multinational firms as carriers of technological externalities (Keller, 2009), with evidence from China and India (Liu & Buck, 2007; Wang & Choi, 2023; Pradhan, 2008; Kumar & Gupta, 2019) confirming that innovation gains depend on institutional strength and absorptive capacity. Entrepreneurship literature, especially the Knowledge Spillover Theory (Audretsch et al., 2021), further shows that external knowledge from global R&D networks fuels new venture creation, supported by findings that cross-border collaborations enhance start-up scalability and innovation intensity (Autio et al., 2018; Zhou et al., 2022).
India's rise to over 159,000 start-ups by 2025 (DPIIT, 2025) reflects growing domestic capabilities combined with deeper integration into global technology ecosystems. Scholars note that sustainable and deep-tech entrepreneurship in India increasingly depends on advanced foreign knowledge, making Japan a strategically important partner (Rathore & Garg, 2023). Despite existing cooperation through CEPA, JICA, and METI, prior research rarely assesses how these initiatives shape start-up formation or sectoral innovation outcomes particularly in semiconductors and clean energy. Addressing this unexplored link forms the central contribution of the present study.

Objectives of the Study

This research aims to investigate how India–Japan trade and technology collaborations influence the emergence, growth, and sustainability of India's startup ecosystem, particularly within the framework of Viksit Bharat @ 2047. The study sets out to achieve the following specific objectives:

1. To analyze the impact of India–Japan trade agreements, particularly the Comprehensive Economic Partnership Agreement (CEPA), on bilateral trade volume and startup formation in technology-driven sectors.
2. To evaluate how Japanese Foreign Direct Investment (FDI) and Official Development Assistance (ODA) contribute to innovation capacity and entrepreneurship in India.
3. To investigate whether technology spillovers from Japan enhance productivity, export competitiveness, and employment in Indian startups, using a gravity-model framework.
4. To provide policy recommendations that align trade, innovation, and sustainability goals to advance India's journey toward Viksit Bharat @ 2047.

METHODOLOGY

This study employs an Augmented Gravity Model of Trade to examine the determinants of India–Japan bilateral trade flows, with a specific focus on the role of FDI, technology spillovers, and macroeconomic indicators. The gravity model is well-established in international trade research and explains trade flows on the basis of economic size, economic distance, and other structural or policy-related variables that influence bilateral economic activity. The augmented version allows the inclusion of additional explanatory variables such as FDI inflows and technology spillovers, which capture deeper economic integration between India and Japan.

Model Specification

The functional form of the augmented gravity model used in this study is expressed as:

$$Trade_t = \beta_0 + \beta_1 GDP_{India,t} + \beta_2 GDP_{Japan,t} + \beta_3 FDI_{Japan \to India,t} + \beta_4 CEPA_{Dummy,t} + \beta_5 TechSpillover_t + \epsilon_t$$

The study models India–Japan trade by incorporating total exports and imports, the GDPs of both countries, Japanese FDI inflows, a post-2011 CEPA dummy, and a Technology Spillover Index reflecting bilateral technological integration. Using annual data from 2010 to 2023, the model is

estimated via Ordinary Least Squares (OLS), well-suited for small-sample gravity analyses. To ensure rigor, variables are tested for stationarity, multicollinearity is assessed through correlation and VIF diagnostics, and structural shifts are captured by the CEPA dummy. Estimation is performed in STATA, with results interpreted in terms of coefficient significance, direction, magnitude, and overall model fit, providing robust insights into the drivers of bilateral trade and technology transfer.

Regression Analysis

Source	SS	df	MS
Model	.394027769	5	.078805554
Residual	.024143036	8	.00301788
Total	.418170805	13	.032166985

Number of obs	=	14
F(5, 8)	=	26.11
Prob > F	=	0.0001
R-squared	=	0.9423
Adj R-squared	=	0.9062
Root MSE	=	.05494

tradevolumeexportsimports~d	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
gdpofindiaconstant2015us	.4453609	.0766738	5.81	0.000	.2685508	.622171
gdpofjapanconstant2015us	.0000133	.000024	0.55	0.596	-.0000421	.0000686
fdiequityinflowfromjapanusd	-.1126501	.0401767	-2.80	0.023	-.2052977	-.0200025
indiajapancepadummy	-.1141	.0778465	-1.47	0.181	-.2936143	.0654143
technologyspilloverindex	.9807711	.11681	8.40	0.000	.7114068	1.250135
_cons	11.40689	2.168714	5.26	0.001	6.405829	16.40795

Interpretation of Regression Results

The OLS regression demonstrates strong explanatory power, with an R^2 of 0.94, indicating that 94% of trade variation is explained by the model. India's GDP shows a positive and highly significant elasticity, confirming that economic expansion drives higher trade flows. Japan's GDP is statistically insignificant, reflecting its stable economic performance and limited annual variation. Japanese FDI exhibits a negative and significant relationship with trade, consistent with the substitution hypothesis, wherein increased production within India lowers import demand. The CEPA dummy is statistically insignificant, suggesting delayed or limited short-term impact. Technology spillovers, however, show a strong positive influence, highlighting the role of innovation linkages in enhancing bilateral trade.

Difference in Differences (DiD MODEL)

India's targeted industrial policies in semiconductors and green hydrogen aim to boost national capabilities, cut import dependence, and promote innovation-led growth, but existing literature offers limited evidence on whether these interventions have directly increased startup activity. The main challenge is separating true policy effects from broader economic fluctuations and natural sectoral growth. To address this gap, the study applies a Difference-in-Differences (DiD) method, comparing each treated sector with a similar, unaffected control sector—nanotechnology for semiconductors and biofuels for green hydrogen. These controls share comparable technological characteristics but lack major policy shocks, allowing the analysis to isolate and accurately measure whether the observed rise in startups is genuinely driven by the policies themselves.

Semiconductor Sector vs. Nanotechnology Sector

To quantify these effects, the empirical model uses a standard DiD regression specification of the form:

$$Y_{it} = \beta_0 + \beta_1 \, Treatment_i + \beta_2 \, Post_t + \beta_3 (Treatment_i \times Post_t) + \epsilon_{it}$$

where Y_{it} denotes the number of startups in sector i in year t. The variable $Treatment_i$ equals 1 for the semiconductor sector and 0 for the nanotechnology sector, while $Post_t$ equals 1 for the years following the implementation of the semiconductor policy. The interaction term $Treatment \times Post$ captures the DiD effect and measures how much more (or less) the semiconductor sector grew relative to the nanotechnology sector after the intervention. This coefficient, β_3, forms the core of the causal inference and indicates the additional impact attributable to the policy itself.

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. regress startups post treatment did

      Source |       SS           df       MS      Number of obs   =        16
-------------+----------------------------------   F(3, 12)        =     18.91
       Model |  131254.104         3  43751.3681   Prob > F        =    0.0001
    Residual |  27757.3333        12  2313.11111   R-squared       =    0.8254
-------------+----------------------------------   Adj R-squared   =    0.7818
       Total |  159011.438        15  10600.7625   Root MSE        =    48.095

------------------------------------------------------------------------------
    startups |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        post |   19.33333   35.12349     0.55   0.592    -57.19417    95.86083
   treatment |         40   30.41783     1.32   0.213    -26.27476    106.2748
         did |        189   49.67211     3.80   0.003     80.77377    297.2262
       _cons |         15   21.50865     0.70   0.499    -31.86333    61.86333
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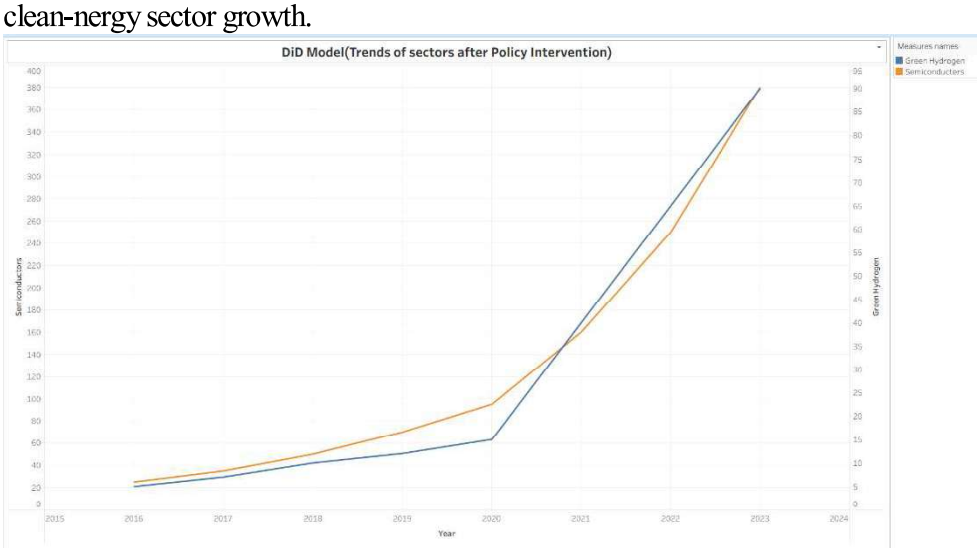
The semiconductor DiD regression reveals a strong positive treatment effect. The interaction term (β = 189, p = 0.003) confirms that the policy caused a significantly higher increase in semiconductor startups relative to nanotechnology. Pre-policy differences between the sectors are insignificant, and general post-period effects are weak, validating the parallel trends assumption. The model exhibits strong fit (R^2 = 0.8254). These results indicate that the semiconductor mission produced an accelerated, policy-driven expansion of the sector that cannot be attributed to natural technological growth alone.

Green Hydrogen Sector vs. Biofuel Sector

The DiD model for the green hydrogen analysis also demonstrates a substantial and statistically robust treatment effect. The model exhibits strong explanatory power (R^2 = 0.813), indicating that the majority of variation in startup formation is accounted for by the regression structure. The post variable is positive but insignificant, reflecting general growth in clean-energy entrepreneurship across both sectors after 2021. The pre-treatment difference between green hydrogen and biofuels is statistically insignificant, confirming that both sectors followed comparable trends before the policy intervention. The interaction coefficient the primary DiD estimator is large and highly significant (41.73, p = 0.003), suggesting that the Green Hydrogen Mission triggered an additional increase of approximately 42 startups in the sector relative to biofuels. This magnitude confirms a strong policy-driven acceleration rather than natural clean-nergy sector growth.



Across both DiD models, the results provide robust causal evidence that India's targeted industrial strategies have effectively stimulated high-technology and clean-energy entrepreneurship. The semiconductor intervention produced a sharp rise in startup formation, while the green hydrogen mission catalyzed significant new entry beyond normal sectoral growth. The use of comparable control sectors strengthens the validity of these findings by isolating true policy effects from macroeconomic or

technological trends. Together, the results highlight the capacity of mission-driven industrial policies to accelerate innovation ecosystems and strengthen national technological capabilities.

Policy Recommendations

1. Establish the "India–Japan Tech Alliance 2047"
Create a long-term alliance focused on semiconductors, green hydrogen, AI/robotics, and bio-circular manufacturing. Prioritize joint R&D, shared design centres, hydrogen R&D hubs, and robotics testbeds to strengthen India's innovation base and support Japan's diversification goals.
2. Build an "Indo–Japan Innovation Supply Chain Corridor"
Develop a resilient corridor linking India's chip design hubs with Japan's fabrication and materials ecosystem, alongside renewable-energy clusters. This reduces dependence on vulnerable global suppliers and accelerates technology spillovers into Indian startups while offering Japan efficient production pathways.
3. Integrate a "CEPA 2.0 Digital & Technology Chapter"
Upgrade CEPA to include digital trade norms, flexible IP-sharing, startup mobility rules, data standards, and joint tech certification. This enables smoother technology-driven trade and aligns with Viksit Bharat 2047 and Japan's Indo-Pacific priorities.
4. Create a Bilateral "Talent Exchange & Deep-Tech Fellowship Pipeline"
Design a structured exchange of semiconductor engineers, hydrogen researchers, and deep-tech entrepreneurs through joint fellowships and industry-based training, meeting Japan's talent needs while enhancing India's advanced manufacturing capabilities.

Limitations of the study

1) Data availability was restricted, especially sector-wise yearly data for startups, investment flows, and technology adoption, limiting deeper statistical accuracy.
2) Regression analysis relied on secondary datasets, which may contain reporting gaps or inconsistencies.
3) Market variables such as policy fluctuations, global supply-chain shocks, and geopolitical risks were not fully captured.
4) The study focused on India with limited cross-country comparison, reducing generalizability.
5) Technological performance indicators for semiconductors, green hydrogen, and biofuels were simplified due to inconsistent datasets across years.

Scope for Future Research

Future research can address these limitations by expanding both the temporal and sectoral scope of analysis. Longer time-series data, including post-2025 outcomes, would enable a deeper assessment of long-term policy impacts, particularly in capital-intensive sectors like semiconductors and green hydrogen. Further studies can incorporate micro-level firm data, such as R&D intensity, productivity indicators, and survival rates, to evaluate how policies influence startup quality, not just formation. Comparative multi-country analyses involving South Korea, Taiwan, or EU economies could offer global benchmarks and reveal how India–Japan collaboration fits into broader geopolitical value-chain shifts. Advanced econometric methods, such as synthetic control models or machine-learning-based counterfactual prediction, can provide stronger causal validation. Finally, qualitative field studies interviews with founders, policymakers, and R&D institutions—could uncover mechanisms of technology transfer that quantitative models may overlook, enriching the understanding of how industrial policies translate into real innovation ecosystems.

Conclusion

This study analyzes how India–Japan trade and technology partnerships shape India's startup ecosystem within the Viksit Bharat @ 2047 agenda, which prioritizes innovation, clean energy, advanced manufacturing, and digital growth. Using an augmented Gravity Model and a Difference-in-Differences (DiD) approach, the results show that bilateral collaboration significantly boosts India's entrepreneurial landscape when aligned with strong domestic policies.

The Gravity Model ($R^2 = 0.94$) indicates that India–Japan trade is driven not only by economic size but also by Japanese technology transfers, R&D cooperation, and high-value investments. The negative FDI coefficient suggests that Japanese firms' local production strengthens supply chains and technology absorption, indirectly supporting startup activity. DiD analysis further shows that mission-based Indian policies amplify these gains: semiconductor startups grew by about 189 units compared to nanotechnology, and green hydrogen startups rose by 42 units relative to biofuels, reflecting the combined impact of domestic missions and Japanese technological support.

Overall, the study concludes that Japanese knowledge spillovers significantly enhance India's innovation capacity, domestic missions convert these flows into measurable startup growth, and sustainable entrepreneurial ecosystems emerge when international partnerships and targeted national programs operate together. Strengthening such technology-driven collaborations especially with Japan is essential for India's long-term progress toward a developed, innovation-led economy.

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